Lab 06: Build a DNS Timeline

An approachable threat-hunting exercise using fictional domain lookups and reserved sample data

NOTEBOOK STYLE Handwriting-inspired typeset notes for printing and revision; not a scan of actual handwriting.

Objective

Build a simple timeline from synthetic DNS records. The goal is to describe what is unusual, compare it with normal sample activity, and record what additional evidence would be needed before making a conclusion.

SYNTHETIC ONLY Every hostname and event is invented for this exercise. The .invalid suffix is reserved for invalid example names, and the example IPs use documentation ranges.

Step 1 - Inspect the fields

Field	Meaning
timestamp_utc	Event time in UTC.
host	Synthetic source endpoint.
dns_query	Fictional queried name.
answer_ip	Reserved documentation address, not a live indicator.
query_count_10m	Teaching count for a short time window.
label	Discussion label, not proof of malicious intent.

Step 2 - Build the timeline

1. Open labs/threat-hunt-dns-samples.csv in a spreadsheet or text editor.
2. Sort by timestamp_utc and group rows by host and dns_query.
3. Compare repeated lookups with the normal queries in the sample.
4. Check whether repeated events are alone sufficient to support the hypothesis; list benign explanations such as application retries.
5. Write a conclusion that states the evidence, uncertainty, and one safe next check.

Step 3 - Ask better questions

Question	Why it matters
Is the hostname itself enough to call this malicious?	No. A name or repetition alone is weak context.
What else was happening on the host at the same time?	Process, sign-in, proxy, and endpoint events may help contextualise it.
Is the system expected to make these queries?	Baseline and application context reduce false positives.

Question	Why it matters
What data is missing?	Query source process, resolver logs, and fuller time context may not be present in the sample.

Deliverable: one-paragraph hunt note

Question: [what pattern did you look for?] Evidence: [synthetic event IDs, times, host, query]. Interpretation: [what the pattern may suggest]. Alternative explanation: [what else could cause it]. Confidence: [low/medium/high and why]. Next step: [additional authorised evidence that would help].

Reference

- MITRE ATT&CK knowledge base
- Microsoft threat hunting overview

SUCCESS CRITERION Describe the pattern without claiming that a training label or repeated DNS lookup proves a compromise.